

and availing the benefits of public and private clouds to store, manage and manipulate data. All data stored in the cloud is encrypted, and cloud services are available under a range of SLAs, covering data integrity and privacy. Many companies have shifted their entire data centres onto the cloud. They enjoy advantages like elasticity, load balancing, redundancy, availability and integrity.

This market is expanding exponentially, currently, with majority of cloud storage being provided by large companies like Google, Microsoft and Amazon. Even companies like Samsung and Apple now have their own cloud storage offerings, such as Samsung Cloud and iCloud, for end users.

Blockchain in cloud storage

A blockchain cloud storage solution takes a user's data and breaks it into small chunks. It then adds an additional layer of security and distributes it throughout the network. This is possible using blockchain features like hashing function, public/private key encryption and transaction ledgers. Each chunk of data is stored in a decentralised location. If intruders try to hack into it, they first get encrypted data and then get only a chunk of data and not the whole file. This secures documents in blockchain-based cloud storage.

Another benefit is that the owner is hidden since the node does not store owner's data. The miner only gets a chunk of data; hence, all sensitive information is protected. Data redundancy and load balancing mechanisms are applied for high availability and quick access. Whenever a user tries to fetch data, all chunks of data are first validated. If any alteration is found on a data chunk, then the miner who altered the data chunk is removed from the network, and that altered part is taken from another redundant copy. Thus, all users get original and identical copies of the data.

Many new companies have jumped onto this blockchain cloud storage market, and the entire business is taking a new shape. Blockchain is the newest and possibly the cheapest way to get cloud storage because many small entities participate in cloud storage by providing their computing power and space to store data. Hence, cloud storage costs are lowered and all entities that participate also get paid.

Companies like Sia, Amazon and Storj are creating systems for paying a small amount to such participants for providing extra storage from their computers. So in the near future, we will see many more cloud storage providers moving to this technology. **EFY**

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